IFD - Innofounder

PROJECT TITLE

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Project title:	Modular development platform for drones and other unmanned systems.
The idea in a nutshell:	An advanced digital development platform for industrial and defense drone manufacturers and operators. The platform uses advanced modeling to let users design, simulate, and optimize UAVs entirely in software - significantly reducing development time and costs.

PREVIOUS APPLICATIONS

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Have you previously applied for a grant from the Innovation Fund with the same company or idea?:	No

CLASSIFICATION

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Political theme:	Critical and digital technologies
Here, you must describe why your idea fits with the chosen theme. If you have chosen, that your idea does not fit within a theme, then you must describe why your idea falls outside the themes. You can only choose one theme.:	The project directly addresses the 'Critical and Digital Technologies' theme by advancing robotics and drone technology, a key area for Denmark's and the EU's strategic autonomy. We provide the digital infrastructure necessary to develop next-generation autonomous systems. By replacing physical trial-and-error with high-fidelity physics simulation, we accelerate innovation in dual-use technologies (logistics, inspection, and security). This strengthens the Danish ecosystem's position in the international technology race by enabling manufacturers to produce more energy-efficient and robust systems faster than global competitors. The platform leverages advanced proprietary algorithms to secure a technological lead in a critical sector.
The project's primary Innovation Area:	Other enginering and technologies
Primary application sector:	Industri

PROJECT DESCRIPTION

ASSESSMENT CRITERIA - QUALITY OF THE IDEA

PLEASE DESCRIBE YOUR BUSINESS IDEA. WHAT PROBLEM, NEED OR DESIRE DOES YOUR BUSINESS IDEA TRY TO MEET?

Helptext:	In this section you should furthermore describe: - What is your solution? - How and why does it work?
Please describe your business idea. What problem, need or desire does your business idea try to meet?:	Industrial drone manufacturers lack reliable digital tools to model and predict system performance, energy consumption, and aerodynamic behaviour during early-stage development. Today, many design decisions are evaluated after building and flight-testing physical prototypes. This hardware-driven development approach leads to long iteration cycles, high engineering costs and performance uncertainty. We are developing a digital engineering platform that enables manufacturers to model, simulate and optimize their drones before physical prototypes are built. By shifting development from empirical testing to specialized high-fidelity simulation, we reduce costly iterations and development time. Our platform is built on a modular architecture, allowing us to easily extend our capabilities to adjacent sectors like rocketry and wind energy. This scalability is driven by direct market demand, and we are already in dialogue with partners requesting these specific features.

PLEASE DESCRIBE HOW YOUR IDEA IS INNOVATIVE

Helptext:	In this section you should furthermore describe: - What is new about your solution? - How it is based on relevant knowledge from your education, research and/or work experience?
Please describe how your idea is	We turnin advanced aerodynamic research into a digital development platform for drones.

innovative:

Existing tools either rely on simplified aerodynamic assumptions or computationally heavy methods that are impractical for iterative engineering. Our solution provides a computationally efficient modelling framework that captures all relevant aspects in one environment.

Since no design fits all missions, drones must be specialized. Our platform drives this continuous optimization of hardware for every unique use case.

The platform is built with a modular architecture, so new capabilities can be added as customer needs evolve, including areas such as mission planning, pilot training and operations monitoring.

The core modelling approach is based on Finn's PhD research in dynamic inflow and multirotor aerodynamics, validated in applied industrial contexts. We build on this by translating research-level modelling into scalable engineering software, driven by validated customer needs.

PLEASE DESCRIBE HOW YOUR BUSINESS IDEA DIFFERS FROM COMPETING SOLUTIONS, PRODUCTS OR SERVICES

Helptext:

In this section you should furthermore describe:
- Who are your competitors, and how do you differ from them?
- How does this difference create value for your customers vis-à-vis competing solutions?
- How does the competition affect your chances of successfully marketing your solution?
- How do you intend to protect your solution from being copied?

Please describe how your business idea differs from competing solutions, products or services:

Current alternatives include generic robotics simulators (e.g. Gazebo), high-end CFD software, and costly "trial-and-error" physical prototyping. Unlike generic tools that use simplified physics, or CFD software that is too slow for rapid iteration, our platform is built exclusively for industrial UAVs. We bridge the gap by providing high-fidelity aerodynamic and energy modeling within a fast, unified digital platform. This disrupts the market by allowing mid-sized manufacturers to out-innovate larger competitors without the R&D overhead, saving the equivalent of a few full-time engineering salaries.

This focus allows engineers to, for example, validate performance and battery life in minutes rather than days, ensuring the final product is perfectly optimized for its specific industrial use case before it is manufactured.

We protect our IP via a SaaS black-box architecture where proprietary models remain secured on our servers. We maintain a lead through continuous integration and extension of our platform to new use cases.

ASSESSMENT CRITERIA - IMPACT

PLEASE RENDER PROBABLE THAT THERE IS A NEED, USERS AND PROBABLY ALSO PAYING CUSTOMERS AND A MARKET OF CONSIDERABLE SIZE - INCLUDING AN INTERNATIONAL ONE, IF POSSIBLE

Helptext:

In this section you should furthermore describe:
- Who are the customers/users, and how do they like your idea?
- Have you sought to identify users and a market for your idea?
- What is the size and potential of this market

Please render probable that there is a need, users and probably also paying customers and a market of considerable size - including an international one, if possible:

Industrial drone manufacturers and drone R&D teams are our primary users, including cargo/logistics drones, inspection and monitoring drones, and specialized component suppliers. Defence and security-related drone applications are also a key focus area, where range, payload, reliability and energy efficiency are mission-critical and where the need for predictable performance is high.

We have validated the need through applied engineering cases and direct dialogue with industry. We already have two Letters of Intent: one from a Danish industrial cargo drone company and one from a Norwegian rotor/wing specialist (~45 employees). The feedback is consistent: companies want to reduce hardware-driven trial-and-error and make better design decisions earlier, before committing time and money to physical prototypes.

The market is international by nature. Danish drone companies compete in this rapidly growing global market. The global drone market is estimated at around USD 70B in 2024 and projected to almost double by 2030.

PLEASE RENDER PROBABLE HOW YOU CAN ESTABLISH A FINANCIALLY SUSTAINABLE BUSINESS BASED ON THE IDEA, AND HOW AN INNOFOUNDER COURSE WILL INCREASE YOUR CHANCES OF DOING SO

Helptext:

In this section you should furthermore describe:
- Specifically how you plan to make money on the idea and to establish a financially sustainable business that can ensure dissemination of your solution.

- The expected business model, revenue structure and pricing.
- How do you expect to disseminate the solution in the market?

Please render probable how you can establish a financially sustainable business based on the idea, and how an Innofounder course will increase your chances of doing so:

We will build a scalable software business with recurring revenue through a subscription platform for industrial drone manufacturers. Drone platforms are continuously redesigned as requirements and technology evolve, creating recurring demand for tools that speed iteration and reduce prototyping costs. The model is tiered (developer and enterprise packages). Enterprise pricing is expected to exceed 100,000 DKK/year, while entry tiers lower adoption barriers.

Costs are mainly software development and ongoing product improvement, which scales well as customers grow. Extra revenue can come from onboarding, training and premium support, while keeping consulting limited.

Go-to-market starts with industrial and defence drone developers where ROI is easiest to document. We will use direct sales, reference cases (two Letters of Intent), and (existing and new) partnerships with component specialists to reach more manufacturers.

Innofounder funds full-time development of v1.0, ROI validation, and preparation for seed funding and international scaling.

PLEASE RENDER PROBABLE HOW YOUR BUSINESS IDEA HAS THE POTENTIAL TO CREATE VALUE FOR DENMARK IN THE FORM OF FINANCIAL GROWTH AND/OR SOLVING SIGNIFICANT SOCIETAL CHALLENGES

Helptext:

In this section you should furthermore describe:
- What value does the solution create for society, e.g. by solving societal challenges, creating growth and employment, etc.?
- Please describe the quantitative and/or qualitative goals of this value creation and be as specific as possible.

Please render probable how your business idea has the potential to create value for Denmark in the form of financial growth and/or solving significant societal challenges:

Danish drone companies compete in a rapidly growing global market. The global drone market is estimated at USD 73.06B in 2024 and projected to reach USD 163.60B by 2030.

Our platform can strengthen Danish drone manufacturers by making development faster, cheaper and more predictable. By enabling simulation and optimization before physical prototyping, companies can reduce costly build–test cycles, improve performance and energy efficiency, and bring new specialized drone platforms to market faster. This improves competitiveness for our customers and supports their international scaling.

Value for Denmark comes from enabling stronger Danish drone companies with higher export potential, as well as creating high-skilled jobs in software and engineering as we build and scale the platform from Denmark. By aiming for an international market, we expect to have the majority of the annual revenue derived from export.

ASSESSMENT CRITERIA - QUALITY OF THE EXECUTION

PLEASE DESCRIBE AND RENDER PROBABLE THAT YOUR TEAM HAS THE COMPETENCES TO REALISE THE IDEA

Helptext:

In this section, you should furthermore describe:
- How does the diversity of the team help you realize the idea?
- What are your abilities to promote and support the idea?
- What insight does the team lack to solve future tasks, and how do you plan to act on this?
- Please state to which extent team members who do not apply for admission will be affiliated to activities relating to the business idea during the Innofounder programme (e.g. state the number of hours per week).

Please describe and render probable that your team has the competences to realise the idea:

Finn is the technical founder. He has written a PhD within the field and has worked on turning the underlying models into practical calculations and approaches. He has tested the methodology through collaboration with companies in our domain and has experienced clear demand for better tools that reduce trial-and-error in drone development.

Flemming is the commercial founder and has worked with sales and marketing for around 20 years. He has sold to large, international companies and has experience building an organisation and driving growth. He is responsible for commercial strategy, partnerships, sales, customer dialogue and later fundraising.

Our combination of strong technical depth and commercial execution gives us a solid starting point to develop the platform, secure the first paying customers and build a scalable company.

We lack deep expertise in SaaS-specific UI/UX design and automated digital onboarding for international scaling. We address this by allocating the development grant to a specialized UI contractor (Activity 1) and will recruit an advisor with SaaS scaling experience to our advisory board during the program.

PLEASE RENDER PROBABLE THAT YOUR BUSINESS IDEA IS REALISTIC AND CAN BE IMPLEMENTED IN PRACTICE

Helptext:

In this section, you should furthermore describe:
- The status of the development and remaining development.
- How far have you come in realizing the idea?
- Which development tasks must be solved during the Innofounder program.
- Please describe all remaining development tasks and be as specific as possible. Which technical, regulatory, and other tasks have you identified, and how do you expect to address them?

Please render probable that your business idea is realistic and can be implemented in practice:

We are not starting from scratch. The core modelling approach has already been developed through Finn's PhD work and has been tested in collaboration with companies in the drone industry. This gives us a solid technical foundation and clear user needs, but it is not yet a product that companies can adopt and pay for.

During the Innofounder programme our main task is to turn the research-based model into a stable and user-friendly platform (v1.0). This includes: packaging the core engine into production-ready software, building a simple interface for configuring drone designs and running simulations, and adding workflows that make results easy to compare and act on. We will validate accuracy and usefulness with our early industry partners and iterate based on real feedback.

On the commercial side we will refine pricing, onboarding and customer support so the platform can be sold as a subscription. Regulatory work is not a primary development blocker, as we do not operate drones ourselves; however, we will ensure the platform outputs are aligned with typical documentation needs for industrial and defence-related drone development, testing and certification.

PLEASE RENDER PROBABLE THAT AN INNOFOUNDER PROGRAM CAN HELP YOU PROPEL YOUR BUSINESS IDEA TO A STAGE WHERE IT IS FINANCIALLY SUSTAINABLE

Helptext:

In this section, you should furthermore describe:
- Where are you when the course ends?
- How do you plan to fund the next step?

Please render probable that an Innofounder program can help you propel your business idea to a stage where it is financially sustainable:

By the end of the Innofounder programme we aim to have a working v1.0 of the platform that customers can use in real development projects, a clear subscription offering and pricing, and the first paying customers on annual subscriptions. We will also have documented results from early users showing reduced build–test iterations and improved decisions on performance and energy use.

The programme gives us the runway to work full-time on product development, validation and go-to-market.

After the programme, the next step will be funded through a combination of subscription revenue and external funding. We expect InnoBooster to be a relevant option to accelerate product development together with pilot customers for next versions. If traction is strong, we will also raise a seed round to scale the team and sales internationally in a high-growth market.

ACTIVITY PLAN

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Activity Number	Activity	Who does it	Result/milestone	Period start	Period end
Activity 1	Engage part-time contractor (100,000 DKK) to build UI and package the platform and integrate it with model pretotype.	Finn + contractor	Initial version of UI and platform v0.5	Month 1	Month 3
Activity 2	Obtaining feedback on v0.5 from new and existing customers on inititial test-projects.	Finn + Flemming	Feedback and roadmap for next development steps required to reach v1.0	Month 3	Month 4
Activity 3	Develop v0.5 into v1.0 based on requirements identified in activity 2.	Finn + contractor	Finished v1.0	Month 5	Month 10
Activity 4	Obtaining feedback on v1.0 from new and existing customers and test the accuracy on real-world scenarios.	Finn	Feedback and roadmap from v1.0, next modeling steps required to obtain model accuracy.	Month 11	Month 12
Activity 5	Pilot customers, development advisory board, in inner circle (LOIs).	Finn + Flemming	Constant feedback to maintain relevance and guide development roadmap.	Month 1	Month 12
Activity 6	Define subscription tiers and pricing; test with pilots and early leads to validate pricing and product market	Flemming	Pricing, packages and draft contracts ready.	Month 1	Month 6

	fit.				
Activity 7	Convert pilots to paid subscriptions.	Finn + Flemming	First paying customers.	Month 10	Month 12
Activity 8	Build distribution via Danish drone ecosystem; prepare InnoBooster + seed pitch.	Flemming	Partner pipeline + pitch deck + funding plan.	Month 2	Month 12

PARTNER DATA

PARTNER MASTER DATA

Is the idea based in a company?: No

FOUNDER 1

First name: Flemming
Last name: Bækdal
Address: Strindbergsvej 19
Zip code: 2500
City: Valby
E-mail address: flemming+finn@forafact.dk
Phone number: 29726117
Country: Denmark
Gender: Male
Citizenship/Nationality: Denmark
Highest completed level of education: Mellemlange videregående uddannelser, MVU
From which educational institution did you graduate?: Aalborg University
Name of education: Bachelor of Science (BSc)
Graduation date or expected date: 15-06-2004
State the precent of ownership of the company for this founder: 50,0%
If necessary, comment on the ownership distribution: We haven't incorporated the company yet

FOUNDER 2

First name: Finn
Last name: Matras
Address: Molbechsvej 14, st. tv.
Zip code: 2500
City: Valby
E-mail address: finn.matras@optim42.com
Phone number: 44189642
Country: Denmark
Gender: Male
Citizenship/Nationality: Germany
Highest completed level of education: Ph.d. og forskeruddannelser
From which educational institution did you graduate?: Other
Name of education: Engineering Cybernetics
Graduation date or expected date: 30-01-2025
State the precent of ownership of the company for this founder: 50,0%

BUDGET

PROJECT

Amount applied for incl. overhead: 760.000 kr.
Monthly fee total: 660.000 kr.
Development costs total: 100.000 kr.

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ATTACHMENTS

PERSONAL INFORMATION

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FOUNDERS

Number of founders: 2

FOUNDER 1

First name: Flemming
Last name: Bækdal
E-mail: flemming+finn@forafact.dk

FOUNDER 2

First name: Finn
Last name: Matras
E-mail: finn.matras@optim42.com

STATEMENTS

APPLICATION FORMALITY

I have read and accepted the terms of use of the electronic application system (set out in the guidelines): ☑

PERSONAL FORMALITIES

I am eligible to apply on behalf of the company: ☑

I / we work full time on the business idea in the course: ☑

I / we do not receive other public grants for what I apply for in Innofounder: ☑

I / we are aware of the requirement for relevant work and residence permits to be able to work on the idea in Denmark in the event that I / we are foreign applicants from EU/EEA countries and countries outside the EU/EEA: ☑

BUSINESS FORMALITIES

A) The applicant is not part of a company structure where an underlying company with a controlling influence is more than 5 years old.: ☑

B) The applicant does not have more than 50 employees: ☑

C) The applicant does not have an annual turnover or balance sheet of more than EUR 10 million in the most recent audited financial statements: ☑

D) The applicant also fulfill A) and B) above, when the applicants companies, which seen in relation to the applicant should be regarded as assosiated or partner companies, as defined the the Comission’s reccomendation concerning the definition of SMEs (2003/361/EF) are included: ☑

E) Public bodies do not control 25% or more of the Applicant's capital and / or voting rights:	☑
F) Applicant is not listed, ie. is not listed on a stock exchange's official list:	☑
G) The applicant was registered with the Danish Business Authority no more than 3 years ago at the time of the application:	☑
H) The applicant has not taken over the activity of another company, which at the time of the transfer either i) formed the basis for turnover with the transferor or ii) could realistically be expected to form the basis for turnover within a period of no more than 36 months:	☑
I) Applicant has not distributed profits:	☑
J) Applicant did not arise through a merger with a company that did not meet point F-I above:	☑
K) The applicant has not received an order for repayment of aid granted by the Danish state and has declared it illegal and incompatible with the Internal Market by the EU Commission, which the applicant has not yet complied with:	☑
L) The applicant is not an undertaking in difficulty regarding the definitions in GBER article 18, litra a-e:	☑